

Q. No.	Question
1.	Explain the difference between a queue and a stack?
2.	Draw a binary Tree for the expression: $A * B - (C + D) * (P / Q)$
3.	Implement PUSH function for stack using array with StackIsFull condition.
4.	Implement POP function for stack using array with StackIsEmpty condition.
5.	Define the Tree Data Structure. List out some applications of Tree-data structure.
6.	Describe binary trees and its types with neat diagram.
7.	List out the steps involved in deleting a node from a binary search tree
8.	Explain tree– traversal and mention the type of traversals?
9.	Create a binary search tree for the following numbers start from an empty binary search tree. 45, 26, 10, 60, 70, 30, 40, 15, 20.
10.	Delete keys 10, 60 and 45 one after the other and show the trees at each stage from the tree constructed in Q. 9.
11.	Construct the recursive solution to tower of Hanoi problems for $n=3$.
12.	Design the algorithm for insertion and deletion in Queue.
13.	Discuss doubly linked list. Explain insertion in doubly linked list.
14.	Explain deletion in doubly linked list.
15.	Consider this infix expression Q: $((A + B) * D) \uparrow (P / Q)$ Translate Q into its equivalent postfix expression P.
16.	Write an algorithm to insert a node in a circular linked list.
17.	Write an algorithm to delete a node from circular linked list.
18.	Explain height, level, depth and path of tree with suitable example.
19.	Explain various operations performed on the Queue?
20.	Explain the usage of stack in recursive algorithm implementation.

Q. No.	Question
1.	Discuss Array with examples. State the difference between an array and a linked list?
2.	Discuss in brief the algorithm to insert a node in a circular linked list.
3.	Discuss an algorithm to delete a node from a linked list.
4.	Explain memory representation of linked lists with diagrams.
5.	Sort the given input array using merge sort. Input Array={ 66,98,34,43,85,87,23,14,98,23 }
6.	Explain a searching problem. List out the steps involved in binary search.
7.	Explain hashing. Discuss the types of collision resolution techniques.
8.	Explain algorithm and complexity of Bubble sort.
9.	Sort the given input array using Selection sort. Input Array={ 66,98,34,43,85,87,23,14,98,23 }

10.	Explain algorithm and complexity of Insertion sort.
11.	Differentiate Selection sort and Insertion sort.
12.	Differentiate Linear search and Binary search algorithm.
13.	Differentiate Merge sort and Quicksort.
14.	List out the steps to search an element in a single linked list.
15.	Design an algorithm to delete a node from the middle(Anywhere but not the first and the last node) of the doubly linked list.
16.	Illustrate all asymptotic notations with suitable graphs.
17.	Differentiate between linear and non linear data structures.
18.	Implement C code for Insertion and Selection sort.
19.	Discuss the advantages of linked lists over arrays.
20.	Differentiate open addressing and closed addressing collision removing technique with example .

Unit-III

Q. No.	Question Statement
1.	Define stack. Implement the operations of stack using arrays.
2.	What is expression? Write the different ways of representing expressions.
3.	Convert the infix expression $a/b - c + d * e - a * c$ into postfix expression. Write a function to evaluate that postfix expression and trace that for given data $a=6, b=3, c = 1, d = 2, e =4$.
4.	Discuss the Tower of Hanoi with example using recursive function.
5.	Explain operations of Stack as ADT with pseudo-code
6.	Give the disadvantage of ordinary queue and how it is solved in circular queue.
7.	Write a note on dequeue.
8.	Define priority queue. Explain in detail the representation of a Priority Queue with example
9.	Define Queue. Implement the operations of queue using arrays.
10.	Write an algorithm to convert infix expression to postfix expression.

UNIT IV

1	Explain Representation methods of binary tree in memory.
2	Explain Traversing of binary tree with appropriate example.
3	Define Tree and following terms with respect to tree with example 1. Root node 2. Siblings 3. Skewed Binary Tree 4. Complete Binary Tree
4	Draw the binary search tree using the following data. Show each step 40 70 35 20 110 75 25 15 73
5	Construct the AVL tree for the following data by inserting each of the following data item one at a time : 10, 20, 15, 12, 25, 30, 14, 22, 35, 40
6	Write a pseudo c++ code for RR, LL, RL, LR rotations
7	Write an Iterative Algorithm for Pre-order and In-order Traversals of binary tree & give suitable example.
8	Explain deletion of node in binary search tree.
9	Write a 'C' function to implement post-order traversal on binary tree
10	Explain threaded binary tree and its representation in link list

Unit-I	
Q. No.	Question Statement
1.	What is a data structure? Give the types of data structures in detail.
2.	What are the basic operations performed on data structures.
3.	Define and explain time and space complexities of an algorithm.
4.	What are different asymptotic notations. Give example for each.
5.	Enlist and explain any 3 sorting techniques.
6.	Explain Insertion Sort and write an algorithm of Insertion Sort
7.	Explain Radix Sort with example
8.	Write an algorithm of selection sort and sort the following numbers using selection sort and show the contents of an array after every pass:- 81, 5, 27, -6, 61, 93, 4, 8, 104, 15
9.	What is hashing? Explain different techniques of hashing.
10.	Explain the quick sort algorithm. Show the contents of array after every iteration of your algorithm start from following status of array.- 27, 76, 17, 9, 57, 90, 45, 100, 79

Unit-II	
1.	Define array. List and explain different operations performed on array.
2.	Write an algorithm to delete element from array.
3.	Write an algorithm to insert an element in array.

4.	Explain Binary Search algorithm with example.
5.	What is Bubble Sort? Write an algorithm of Bubble Sort.
6.	Explain in brief the different searching techniques. What is the time complexity of each of them?
7.	Define Linked List. Write the advantages and disadvantages of Linked List.
8.	Write an pseudo code for inserting element at the end of the list.
9.	Write a short note on Circular Link List. Give its applications.
10.	What is Two-way List. Give its advantages and disadvantages.